Grade 10 Practice

Standard G.CO.1

Name: _____ Date: _____

Computation

1. How many dimensions does a point have? _____ **0**

2. How many directions does a line extend? _____ **2**

3. An angle bisector divides an angle into how many equal angles? _____ **2**

4. How many endpoints does a segment have? _____ **2**

Word problems

5. How many endpoints does a ray have?

Show your work and answer.

Answer: 1

6. Two lines in the same plane that never meet are called what?

Show your work and answer.

Answer: parallel

Grade 10 Practice

Standard G.CO.6

Name: _____ Date: _____

Computation

1. Does a reflection preserve distance? Write yes or no. **yes**

2. Does a dilation preserve angle measures? Write yes or no. **yes**

3. Does a dilation preserve distance? Write yes or no. **no**

4. Is a figure congruent to its rotation? Write yes or no. **yes**

Word problems

5. A triangle is reflected across a line. Is the image congruent to the original? Write yes or no.

Show your work and answer.

Answer: yes

6. A triangle is dilated by a factor of 2. Is the image congruent to the original? Write yes or no.

Show your work and answer.

Answer: no

Grade 10 Practice

Standard G.CO.7

Name: _____ Date: _____

Computation

1. If all three pairs of corresponding sides are equal, are the triangles congruent? Write yes or no.

yes

2. If two sides and the included angle are equal, are the triangles congruent by SAS? Write yes or no.

yes

3. If only two angles are equal, are the triangles congruent? Write yes or no.

no

4. Is SSA always enough to prove congruence? Write yes or no.

no

Word problems

5. Triangles have sides 3, 4, and 5 and another has sides 3, 4, and 5. Are they congruent? Write yes or no.

Show your work and answer.

Answer: **yes**

6. Two triangles have two equal angles and one equal side. Are they congruent by AAS? Write yes or no.

Show your work and answer.

Answer: **yes**

Grade 10 Practice

Standard G.CO.8

Name: _____ Date: _____

Computation

1. Two triangles have all three sides equal. Name the criterion: _____

SSS

2. Two triangles have two sides and the included angle equal. Name the criterion: _____

SAS

3. Two triangles have two angles and the included side equal. Name the criterion: _____

ASA

4. Two right triangles have equal hypotenuses and one equal leg. Name the criterion: _____

HL

Word problems

5. Two triangles share a side and have two pairs of equal angles. Which criterion applies, ASA or AAS?

Show your work and answer.

Answer: ASA or AAS

6. Two right triangles have equal hypotenuses and one equal leg. Name the criterion.

Show your work and answer.

Answer: HL

Grade 10 Practice

Standard G.CO.9

Name: _____ Date: _____

Computation

1. Vertical angles are equal. Write yes or no.

yes

2. Angles on a straight line add up to how many degrees? _____

180

3. When two parallel lines are cut by a transversal, alternate interior angles are equal. Write yes or no.

yes

4. The angles of a triangle add up to how many degrees? _____

180

Word problems

5. Angles around a point add up to how many degrees?

Show your work and answer.

Answer: 360 degrees

6. Corresponding angles are equal when the lines are what?

Show your work and answer.

Answer: parallel

Grade 10 Practice

Standard G.CO.10

Name: _____ Date: _____

Computation

1. In an isosceles triangle, the base angles are _____ equal.

equal

2. A midsegment of a triangle is _____ the length of the third side.

half

3. A triangle has angles 50° and 60°. What is the third angle? _____

70°

4. An exterior angle equals the sum of two remote interior angles of 60° and 50°. What is the exterior angle? _____

110°

Word problems

5. In an isosceles triangle, one base angle is 40°. What is the vertex angle?

Show your work and answer.

Answer: **100°**

6. A triangle has a base of 10. What is the length of its midsegment parallel to the base?

Show your work and answer.

Answer: **5**

Grade 10 Practice

Standard G.CO.11

Name: _____ Date: _____

Computation

1. Opposite sides of a parallelogram are equal. Write yes or no.

yes

2. Diagonals of a rectangle are equal. Write yes or no.

yes

3. Diagonals of a rhombus are perpendicular. Write yes or no.

yes

4. Consecutive angles of a parallelogram add up to how many degrees?

_____

180

Word problems

5. A parallelogram has an angle of 70°. What is an adjacent angle?

Show your work and answer.

Answer: **110°**

6. In a rectangle, the diagonals bisect each other. Do their midpoints match? Write yes or no.

Show your work and answer.

Answer: **yes**

Grade 10 Practice

Standard G.SRT.2

Name: _____ Date: _____

Computation

1. Two triangles have two pairs of equal angles. Are they similar? Write yes or no.

yes

2. A triangle has a scale factor of 3 from another. What is the ratio of their perimeters? _____

3

3. Triangles with angles 40° and 60° and 40° and 80° — are they similar? Write yes or no.

no

4. Triangles with side ratios 1:2 and 2:4 — are they similar? Write yes or no.

yes

Word problems

5. Triangles have sides 6, 8, and 10, and others have sides 3, 4, and 5. Are they similar? Write yes or no.

Show your work and answer.

Answer: yes

6. A triangle has a side of 5 and is dilated by a scale factor of 2. What is the new side length?

Show your work and answer.

Answer: 10

Grade 10 Practice

Standard G.SRT.4

Name: _____ Date: _____

Computation

1. Triangle with sides 3, 4, and 5. Is it a right triangle? Write yes or no.

yes

2. Triangle with sides 5, 12, and 13. Is it a right triangle? Write yes or no.

yes

3. Triangle with sides 4, 5, and 6. Is it a right triangle? Write yes or no.

no

4. Two triangles have two equal angles. What theorem makes them similar? Write AA.

AA

Word problems

5. A ladder's shadow and a tree's shadow form similar triangles. What theorem proves it when two angles match?

Show your work and answer.

Answer: AA

6. Two right triangles share one acute angle. Are they similar? Write yes or no.

Show your work and answer.

Answer: yes

Grade 10 Practice

Standard G.SRT.5

Name: _____ Date: _____

Computation

1. Similar triangles have sides 3 and 6. A side of 4 in the smaller triangle matches what side in the larger? _____

8

2. Ratio of similar triangles is 2:5. A side of 10 in the smaller matches what side in the larger? _____

25

3. A 2 m post casts a 1 m shadow when a 10 m tree casts a 5 m shadow. How does this confirm similarity?

ratios match

4. Scale factor 1.5 applied to side 8. What is the new side? _____

12

Word problems

5. A 6 ft person casts a 4 ft shadow at the same time a tree casts a 20 ft shadow. How tall is the tree?

Show your work and answer.

Answer: 30 feet

6. Two similar triangles have sides 4 and 6. If the smaller triangle has a side of 10 that matches the 4, what is the matching side in the larger?

Show your work and answer.

Answer: 15

Grade 10 Practice

Standard G.SRT.6

Name: _____ Date: _____

Computation

1. In a right triangle, sin = opposite / _____ **hypotenuse**

2. Find cos 60°. _____ $\frac{1}{2}$

3. Find tan 45°. _____ **1**

4. Find sin 90°. _____ **1**

Word problems

5. A right triangle has opposite side 3 and adjacent side 4. What is tan A?

Show your work and answer.

Answer: $\frac{3}{4}$

6. A right triangle has hypotenuse 10 and opposite side 6. What is sin A?

Show your work and answer.

Answer: $\frac{3}{5}$

Grade 10 Practice

Standard G.SRT.8

Name: _____ Date: _____

Computation

1. A right triangle has legs 3 and 4. Find the hypotenuse. _____

5

2. A right triangle has hypotenuse 10 and angle 30°. Find the opposite side. _____

5

3. A right triangle has legs 5 and 12. Find the hypotenuse. _____

13

4. A right triangle has adjacent side 6 and angle 45°. Find the opposite side. _____

6

Word problems

5. A ramp is 10 m long at an angle of 15°. About how high is it? Round to the nearest hundredth.

Show your work and answer.

Answer: **2.59 m**

6. A ladder 13 ft long reaches a wall 12 ft high. How far is its base from the wall?

Show your work and answer.

Answer: **5 ft**

Grade 10 Practice

Standard G.C.2

Name: _____ Date: _____

Computation

1. A central angle measures 80°. What is the measure of its arc? _____ **80°**

2. An inscribed angle is _____ its intercepted arc. Write half or double. **half**

3. An inscribed angle intercepts a 100° arc. What is the inscribed angle? _____ **50°**

4. An inscribed angle intercepts a semicircle. What is its measure? _____ **90°**

Word problems

5. A central angle is 120°. What is an inscribed angle on the same arc?

Show your work and answer.

Answer: 60°

6. A central angle is 70°. What is an inscribed angle on the same arc?

Show your work and answer.

Answer: 35°

Grade 10 Practice

Standard G.C.5

Name: _____ Date: _____

Computation

1. A circle has radius 6 and a 60° central angle. What is the arc length? Write in terms of π.

2π

2. A sector has radius 6 and a 90° angle. What is its area? Write in terms of π.

9π

3. A circle has radius 3. What is its circumference? Write in terms of π.

6π

4. A circle has radius 10 and a 36° arc. What is the arc length? Write in terms of π.

2π

Word problems

5. A pizza has radius 8 in. A $\frac{1}{4}$ slice is cut. What is the arc length of the crust? Write in terms of π.

Show your work and answer.

Answer: 4π inches

6. A sector has radius 4 and a 180° angle. What is its area? Write in terms of π.

Show your work and answer.

Answer: 8π

Grade 10 Practice

Standard G.GPE.4

Name: _____ Date: _____

Computation

1. What is the midpoint of (0, 0) and (4, 6)? _____ **(2, 3)**

2. What is the distance from (0, 0) to (3, 4)? _____ **5**

3. What is the slope of the line from (0, 0) to (3, 4)? _____ $\frac{4}{3}$

4. Is the point (1, 2) on the line y = 2x? Write yes or no. **yes**

Word problems

5. Parallelogram ABCD has segments AB and CD with equal slopes. Are they parallel? Write yes or no.

Show your work and answer.

Answer: parallel

6. What is the midpoint of (1, 1) and (3, 5)?

Show your work and answer.

Answer: (2, 3)

Grade 10 Practice

Standard G.GPE.6

Name: _____ Date: _____

Computation

1. What point divides (0, 0) to (4, 8) into a 1:1 ratio? _____ **(2, 4)**

2. What point divides (0, 0) to (6, 0) into a 1:2 ratio? _____ **(2, 0)**

3. What is the midpoint of (-2, 4) and (2, 8)? _____ **(0, 6)**

4. What point is $\frac{1}{4}$ of the way from (0, 0) to (8, 0)? _____ **(2, 0)**

Word problems

5. What is the midpoint of the segment from (1, 1) to (5, 9)?

Show your work and answer.

Answer: (3, 5)

6. What point is $\frac{2}{3}$ of the way from (0, 0) to (9, 0)?

Show your work and answer.

Answer: (6, 0)

Grade 10 Practice

Standard G.GPE.7

Name: _____ Date: _____

Computation

1. A rectangle has corners (0, 0), (5, 0), (5, 3), and (0, 3). What is its area? _____

15

2. A right triangle has vertices (0, 0), (4, 0), and (0, 3). What is its perimeter? _____

12

3. What is the distance from (0, 0) to (5, 12)? _____

13

4. A square has side 6 units. What is its perimeter? _____

24

Word problems

5. A rectangle has corners (1, 2), (7, 2), (7, 6), and (1, 6). What is its perimeter?

Show your work and answer.

Answer: 20 units

6. A triangle has vertices (0, 0), (6, 0), and (3, 4). What is its area?

Show your work and answer.

Answer: 12 square units

Grade 10 Practice

Standard G.GMD.1

Name: _____ Date: _____

Computation

1. What is the circumference of a circle with radius 5? Write in terms of π.

10π

2. What is the area of a circle with radius 4? Write in terms of π.

16π

3. What is the area of a circle with diameter 6? Write in terms of π.

9π

4. What is the circumference of a circle with diameter 8? Write in terms of π.

8π

Word problems

5. A tree trunk has a diameter of 10 feet. What is its circumference? Write in terms of π.

Show your work and answer.

Answer: 10π feet

6. A circular rug has radius 3 feet. What is its area? Write in terms of π.

Show your work and answer.

Answer: 9π square feet

Grade 10 Practice

Standard G.GMD.3

Name: _____ Date: _____

Computation

1. What is the volume of a cylinder with radius 2 and height 6? Write in terms of π.

24π

2. What is the volume of a cone with radius 3 and height 6? Write in terms of π.

18π

3. What is the volume of a sphere with radius 3? Write in terms of π.

36π

4. What is the volume of a cylinder with radius 1 and height 10? Write in terms of π.

10π

Word problems

5. A can has radius 4 cm and height 10 cm. What is its volume? Write in terms of π.

Show your work and answer.

Answer: 160π cubic cm

6. A cone has radius 3 cm and height 4 cm. What is its volume? Write in terms of π.

Show your work and answer.

Answer: 12π cubic cm

Grade 10 Practice

Standard G.MG.1

Name: _____ Date: _____

Computation

1. A can is modeled as a cylinder. What shape is the model? _____

cylinder

2. A tent is modeled as a triangular prism. Name the three dimensions needed to find its volume.

base, height, and length

3. A box is modeled as a rectangular prism 2 by 3 by 4. What is its volume?

24

4. A cone is modeled with radius 2 and height 9. What is its volume? Write in terms of π.

12π

Word problems

5. A swimming pool is modeled as a rectangular prism 20 ft by 10 ft by 4 ft. What is its volume in cubic feet?

Show your work and answer.

Answer: 800 cubic feet

6. A grain silo is modeled as a cylinder with radius 5 and height 20. What is its volume? Write in terms of π.

Show your work and answer.

Answer: 500π cubic units